# Math 5050 – Special Topics: Algebraic Topology– Spring 2026

w/Professor Shibo Liu

## Notes and Observations

Notes and Observations

---

### Helpful Diagrams

### Topological Objects

- $n$-Ball: unit ball in $\mathbb{R}^n$.
  Line (1-ball): $D^1 = \{x : x^2 \leq 1\}$
  Disk (2-ball): $D^2 = \{(x, y) : x^2 + y^2 \leq 1\}$
  $n$-ball: $D^n = \{(x_1, \ldots, x_n) : \sum x_i^2 \leq 1\}$
  '$e$-cell': interior of $n$-ball, $e^n = \{(x_1, \ldots, x_n) : \sum x_i^2 < 1\}$
- $n-1$-surface: $S^{n-1} = \partial D^n = \{(x_1, \ldots, x_n) : \sum x_i^2 = 1\}$
  Two-point: $S^0 = \partial D^1 = \{0, 1\}$.
  Cirlce: $S^1 = \partial D^2 = \{(x, y) : x^2 + y^2 = 1\}$.
  Sphere: $S^2 = \partial D^3 = \{(x, y, z) : x^2 + y^2 + z^2 = 1\}$.
- A ***cone*** on $X$ is the quotient space $cX = (X \times I)/(X \times \{1\})$.
  Every element in $cX$ is $(X, t)$ where $0 \leq t \leq 1$ and $(X, 1)$ is the identity element.

### Invariant Topological Properties

- Connectedness.
- Contractibility.
- Compactness.
- Completeness.

### Operators on topological objects

1. **Product** $X \times Y$. For every point in $X$ we 'attach' a copy of $Y$.
   Boundary of Product: $\partial(X \times Y) = (\partial X \times Y) \cup (X \times \partial Y)$
   Beware $S^1 \times S^1$ can describe sphere as well as a torus but they aren't isomorphic.
2. **Quotient** $Y/X$
   Take $X$ to be a point and map all of $Y$ to it.
3. **Suspensions**
4. **Join**
5. **Wedge Sum**
6. **Smash Product**

## Vocabulary from Topology

- **Properly Discontinuous**

  A properly discontinuous action of a group $G$ on a topological space $X$ is a group action where for every point $x \in X$, there exists a neighborhood $U$ such that the set of group elements $g \in G$ moving $U$ to intersect itself is limited to the identity, i.e., $gU \cap U \neq \emptyset$ only for $g = e$. This ensures the quotient map $X \to X/G$ is a covering map.

- **Connected**

  A topological space $X$ is said to be *disconnected* if it is the union of two disjoint non-empty open sets. Otherwise, $X$ is said to be ***connected.***

- **Simply Connected**

  A topological space is simply connected if it is path-connected and every continuous loop within it can be continuously shrunk to a point (i.e., trivial group).

- **Path Connected**

  A path from a point $x$ to a point $y$ in a topological space $X$ is a continuous function $f$ from the unit interval $[0, 1]$ to $X$ with $f(0) =$ and $f(1) = y$. A path-component of $X$ is an equivalence class of $X$ under the equivalence relation which makes $x$ equivalent to $y$ if and only if there is a path from $x$ to $y$. The space $X$ is said to be path-connected (or pathwise connected or **0**-connected) if there is exactly one path-component.

- **Locally Path Connected**

  A space $X$ is called locally connected at $x$ if every neighborhood of $x$ contains a connected open neighborhood of $x$, that is, if the point $x$ has a neighborhood base consisting of connected open sets.

- **Semi-locally Path Connected**

  A space $X$ is called semi-locally simply connected if every point $x$ in $X$ and every neighborhood V of x has an open neighborhood $U$ of x such that $x \in U \subset V$ with the property that every loop in $U$ can be contracted to a single point within $X$ (i.e. every loop in $U$ is nullhomotopic in $X$).

- **Homeomorphic**

  Give two topological spaces $X$ and $Y$. They are said to be ***homeomorphic*** if there exists a continuous function $f : X \to Y$ and its inverse. (think "isomorphsim" but in topological terms).

- **Locally Homeomorphic**

- **Quotient Topology**

## Deformation Retract

**Definition 0.0.1** (Deformation Retract)**.** A subspace $A$ of $X$ is ***deformation retract*** of $X$ if there is a retraction $g : X \to A$ (i.e., a map $g : X \to A$ with $g(a) = a$ for every $a \in A$) such that $g$ is homotopic to the identiy map $id : X \to X$. A subspace $A$ of $X$ is a ***strong deformation retract*** if there is a retraction $g : X \to A$ such that $g$ is homotopic rel $A$ to the identity map $id : X \to X$.

## Homotopy Equivalence and Deformation Retract

***Homotopy***: Two maps $f_0 : (X, A) \to (Y, B)$ and $f_1 : (X, A) \to (Y, B)$ are ***homotopic***, i.e., $f_0 \sim f_1$, if there is a map

$$F : (X, A) \times I \to (Y, B)$$

where $F(x, 0) = f_0$ and $F(x, 1) = f_1$ for every $x \in X$. *Homotopy* is an equivalence relation.

**Definition 0.0.2** (Homotopy Equivalence)**.** A continuous map $f : X \to Y$ is a ***homotopy equivalence*** if there exists a continuous map $g : Y \to X$ such that $f \circ g = \mathbb{I}_Y$ and $g \circ f = \mathbb{I}_X$. This means the maps are inverses of each other up to homotopy, implying that the spaces $X$ and $Y$ share the same homotopy type.

**Theorem 0.0.3.** *Put simply two spaces are homotopy equivalent if there is a third space that deformation retracts (shrinks/expands) to both.*

**Lemma 0.0.4.** *A cell complex, $X$ and a subcomplex $A$, then the quotient map $X \to X/A$ is a homotopy equivalence.*

**Theorem 0.0.5** (Homeomorphism implies Isomorphism (not visa versa))**.** *If $X$ is homeomorphic to $Y$ then $\pi_1(X) \cong \pi_1(Y)$.*

**Theorem 0.0.6** (Homotopy Equivalence implies Isomorphism)**.**

*Remark* 0.0.7. Understand a progression of relationships between two topological spaces $X, Y$ from"isomorphic/homeomorphic" $\to$ "deformation retract between $X$ and $Y$" $\to$ "homotopy equivalent".

**Further:** $X$ and $Y$ need only be a deformation retract between them.

### The Fundamental Group

The ***Fundamental Group*** is the set of homotopy classes of loops.

**Theorem 0.0.8** (The Fundamental Group is a group)**.** .

*The Fundamental Group $\pi_1(X, x_0)$ is a group*

**Theorem 0.0.9.** .

$\pi_1(S^1) \cong \mathbb{Z}$[1]

*Proof: Let $a$ be the generator of $\pi_1(S^1)$, i.e., $\pi_1(S^1) = \langle a \rangle$. Define $\varphi : \mathbb{Z} \to \pi_1(S^1)$ as*

$$\varphi(n) = a^n$$

*Then*

- *$\varphi$ is surjective, i.e., $\varphi(m) = \varphi(n) \implies m = n$.*
- *$\varphi$ is injective, $\forall [f] \in S^1 \, \exists n \in \mathbb{Z} \to \varphi(n) = a^n = f$*

*Thus, an isomorphism exists between $\mathbb{Z}$ and $\pi_1(S^1)$*

*Proof (outline): Show that there exists a homomorphism $\varphi : \mathbb{Z} \to \pi_1(S^1)$ then show that*

- *$\varphi$ is surjective, i.e., $\varphi(m) = \varphi(n) \implies m = n$.*
- *$\varphi$ is injective, $\forall [f] \in S^1 \, \exists n \in \mathbb{Z} \to \varphi(n) = f$*

*Remark* 0.0.10. In the video *Algebraic Topolgy 3: The Fundamental Group is a Group*, the instructor uses a helix to describe multiple but separate paths around $S^1$ indicating that a full loop in $S^1$ can be a path on the helix from a point to the same point on the next (or successive) level. The helix is eventually mapped to $\mathbb{R}$ where the closed loops are then mapped to $\mathbb{Z}$.

### Covering Space

If $\tilde{X}$ is a covering space for $X$ then $\pi_1(\tilde{X})$ is a subgroup of $\pi_1(X)$. And there is a bijection of the set/space of maps $\tilde{X} \to X$ and subgroups of $\pi_1(X)$.

A symmetric foundamental group is normal

[1] Note that this is independent of base-point $x_0$.

**Definition 0.0.11** (Covering Space)**.**

$$\begin{aligned} p : \tilde{X} &\to x \\ \tilde{x_0} &\mapsto x_0 \text{ basepoint } (X, x_0) \end{aligned}$$

open covers $\{u_\alpha\}$ of $X$ such that $p^{-1}(\alpha)$ is homeomorphic to $u_\alpha$.

There is an induced mapping $p_* : \pi_1(\tilde{X}, \tilde{X}_0) \to \pi_1(X, x_0)$ where $p_*$ is a subgroup of $\pi_1(X, x_0)$.

*Remark* 0.0.12. It is the projection, $p$, along with the preimage homeomorphism that forms this connection between Covering Spaces and Fundamental Groups.

# Simplicial Diagrams

### Torus

$$\begin{tikzcd} v \arrow[r, "b"] & v \\ v \arrow[u, "a"] \arrow[r, "b"'] \arrow[ur, "c"] & v \arrow[u, "a"'] \end{tikzcd}$$

# Homology

### Simplicial Mapping, $\Delta^n$

The ***simplicial mapping***, $\Delta_n(X)$, is the set of all mappings of n-simplexes onto the topological space $X$. These mappings form a group.

### Chain Complex

Given a set of nested groups $C_n$ such that

$$\cdots \to C_{n+1} \xrightarrow{\partial_{n+1}} C_n \xrightarrow{\partial_n} C_{n-1} \xrightarrow{\partial_{n-1}} \cdots \longrightarrow C_1 \xrightarrow{\partial_1} C_0 \xrightarrow{\partial_0} \to 0$$

Where the image of $\operatorname{img} \partial_{n+1} \subset \ker \partial_n$ for all $n$. We define the ***exact chain*** if $\operatorname{img} \partial_{n+1} = \ker \partial_n$ for all $n$.

### The $n^{\text{th}}$ simplicial homology group, $H_n^{\Delta}(X)$

The chain complex where $C_n = \Delta_n$ of simplicial elements. Where each element is defined as $H_n(X) = \ker \partial_n / \operatorname{img} \partial_{n+1}$.

### The $n^{\text{th}}$ singular homology group, $H_n(X)$

The chain complex where $C_n$ is the set of continuous functions on $X$ in $n$ variables. Where each element is defined as $H_n(X) = \ker \partial_n / \operatorname{img} \partial_{n+1}$.

# Exact Chain Complexes

### Small Exact Chains

Define $i : A \to B$ and $j : B \to C$ and

$$0 \longrightarrow A \xrightarrow{\quad i \quad} B \xrightarrow{\quad j \quad} C \longrightarrow 0$$

$$\ker(i) = 0 \longrightarrow \operatorname{img}(i) = \ker(j) \longrightarrow \operatorname{img}(j) = C$$

Question: $i \circ j : A \to C$ is injective and $(j \circ i)^{-1} : C \to A$ is surjective?

**Applied to Chain Complexes**

# AI Questions:

### What are the properties of Covering Spaces?

A covering space is a concept deeply connected to the fundamental group. Let's explore its properties.

Let $p : E \to B$ be a continuous surjective map between topological spaces. We call $p$ a ***covering map***, and $E$ a ***covering space*** of $B$, if it satisfies the following property:

For every point $b \in B$, there exists an open neighborhood $U$ of $b$ in $B$ such that its preimage $p^{-1}(U)$ is a disjoint union of open sets $V_i$ in $E$, and for each $i$, the restriction of $p$ to $V_i$, denoted $p|_{V_i} : V_i \to U$, is a homeomorphism. The neighborhood $U$ is called an ***evenly covered*** neighborhood.

**Key Properties of Covering Spaces:**

1. **Local Homeomorphism:** A covering map $p : E \to B$ is a local homeomorphism. This means that for every point $e \in E$, there is a neighborhood of $e$ that is mapped homeomorphically onto its image in $B$. This follows directly from the definition of an evenly covered neighborhood.
2. **Lifting Properties:** Covering spaces have unique lifting properties for paths and homotopies. This is where the connection to the fundamental group becomes crucial.
   - **Path Lifting Property:** Given a path $\gamma : [0, 1] \to B$ in the base space and a starting point $e_0 \in E$ such that $p(e_0) = \gamma(0)$, there exists a unique path $\tilde{\gamma} : [0, 1] \to E$ in the covering space that starts at $e_0$ (i.e., $\tilde{\gamma}(0) = e_0$) and covers $\gamma$ (i.e., $p \circ \tilde{\gamma} = \gamma$). This lifted path $\tilde{\gamma}$ is called a ***lift*** of $\gamma$.
   - **Homotopy Lifting Property:** If two paths in $B$ are homotopic, their lifts in $E$ starting at the same point will also be homotopic (specifically, they will end at the same point).
3. **Relationship with the Fundamental Group:** The connection between covering spaces and the fundamental group is one of the most important results in algebraic topology.
   - Let $p : (E, e_0) \to (B, b_0)$ be a covering map. The induced homomorphism $p_* : \pi_1(E, e_0) \to \pi_1(B, b_0)$ is injective. This means the fundamental group of the covering space is isomorphic to a subgroup of the fundamental group of the base space.
   - The set of lifts of a loop in $B$ based at $b_0$ corresponds to the elements of the subgroup $p_*(\pi_1(E, e_0))$. Specifically, a loop $\gamma$ in $B$ lifts to a loop in $E$ if and only if its homotopy class $[\gamma]$ is in the subgroup $p_*(\pi_1(E, e_0))$.
4. **Classification of Covering Spaces:** For a 'nice' space $B$ (path-connected, locally path-connected, and semilocally simply-connected), there is a one-to-one correspondence between the subgroups of its fundamental group $\pi_1(B, b_0)$ and the isomorphism classes of based covering spaces of $B$. The trivial subgroup $\{e\}$ corresponds to a special covering space called the **universal covering space**, which is simply connected (i.e., its fundamental group is trivial).
5. **From another query: How is a Covering Space Related to a subgroup of a Fundamental Group?**. The terminology is a misnomer. The word 'covering' implies 'larger than' as in the definition of 'compact' has 'finite subcover' and whose union is larger. A covering space is much more related to a subgroup which IS NOT be as large as the original group.

**Describe the homology of a torus.**

To describe the homology of a torus, we consider the standard 2-torus, denoted as $T^2$ or $S^1 \times S^1$. The homology groups $H_n(T^2; \mathbb{Z})$ provide information about the "holes" of various dimensions within the shape.

$$\begin{tikzcd} v \arrow[r, "b"] & v \\ v \arrow[u, "a"] \arrow[r, "b"'] & v \arrow[u, "a"'] \end{tikzcd}$$

**Step-by-Step Explanation**

1. **Definition and CW Structure**
   A torus can be constructed by taking a square and identifying opposite edges. This gives it a CW complex structure consisting of:
   - **One 0-cell (vertex):** Let's call it $v$.
   - **Two 1-cells (edges):** Let's call them $a$ and $b$. Both start and end at $v$.
   - **One 2-cell (face):** Let's call it $f$. The boundary of this face is attached along the path $aba^{-1}b^{-1}$ (the commutator of the two edges).
2. **Calculating $H_0(T^2)$ (Connectedness)**
   The $0^{\text{th}}$ homology group represents the number of path-connected components. Since the torus is a single connected piece, the $0^{\text{th}}$ homology group is the free abelian group of rank 1.
   **Result:** $H_0(T^2) \cong \mathbb{Z}$.
3. **Calculating $H_1(T^2)$ (1-Dimensional Holes)**
   The 1st homology group represents the "loops" that cannot be shrunk to a point or expressed as boundaries of a surface.
   - From the CW structure, the boundary of the 2-cell $f$ is $\partial f = a + b - a - b = 0$ in the abelian chain group.
   - Since the boundary is zero, the two 1-cells $a$ and $b$ are cycles and are not boundaries of any 2-cell.
   - These two cycles correspond to the "meridional" and "longitudinal" circles of the torus.
   - **Result:** $H_1(T^2) \cong \mathbb{Z} \oplus \mathbb{Z}$ (or $\mathbb{Z}^2$).
4. **Calculating $H_2(T^2)$ (2-Dimensional Voids)**
   The 2nd homology group represents enclosed volumes or "cavities."
   - The torus is a compact, connected, oriented 2-manifold without boundary.
   - For any such manifold of dimension $n$, the top homology group $H_n$ is isomorphic to $\mathbb{Z}$.
   - In our CW complex, the single 2-cell $f$ has a boundary of 0, making it a cycle. There are no 3-cells to serve as boundaries for it.

   * **Result:** $H_2(T^2) \cong \mathbb{Z}$.
5. **Higher Homology Groups**
   Since the torus is a 2-dimensional manifold, there are no cells of dimension 3 or higher.
   * **Result:** $H_n(T^2) = 0$ for all $n \geq 3$.

**Final Answer Summary**

The homology groups of the torus $T^2$ with integer coefficients are summarized in the table below:

— Dimension ($n$) — Homology Group $H_n(T^2;\mathbb{Z})$ — Description — ——————— — 0 — $\mathbb{Z}$ — The torus is path-connected. — — 1 — $\mathbb{Z}\oplus\mathbb{Z}$ — Two independent non-contractible loops (meridian and longitude). — — 2 — $\mathbb{Z}$ — The torus encloses a single 2-dimensional "void" (it is an oriented surface). — — $\geq 3$ — 0 — No higher-dimensional features exist. —

Final Answer: Describing the homology of a torus Defining the CW complex structure To analyze the torus $T^2$, we can view it as a square with opposite sides identified. This construction allows us to define a CW complex structure, which simplifies the calculation of homology groups. The structure consists of:

- - **One 0-cell (vertex):** A single point $v$ where all corners of the square meet.
- - **Two 1-cells (edges):** Two loops, $a$ and $b$, that start and end at $v$.
- - **One 2-cell (face):** The interior of the square, $f$, attached to the edges along the boundary path $aba^{-1}b^{-1}$.

Calculating the $0^{\text{th}}$ homology group $H_0$ The $0^{\text{th}}$ homology group $H_0(T^2;\mathbb{Z})$ measures the path-connectedness of the space. Because the torus is a single, continuous piece, it has only one connected component. In algebraic terms, this corresponds to a free abelian group of rank 1.

$$H_0(T^2) \cong \mathbb{Z}$$

Calculating the 1st homology group $H_1$ The 1st homology group $H_1(T^2;\mathbb{Z})$ identifies 1-dimensional "holes" or loops that cannot be contracted to a point. In our CW structure, the boundary of the 2-cell $f$ is calculated as:

$$\begin{aligned}\partial f &= a + b - a - b \\ &= 0\end{aligned}$$

Since the boundary is zero, the loops $a$ (the meridian) and $b$ (the longitude) are cycles that do not bound any surface. They are independent, leading to a rank-2 free abelian group:

$$H_1(T^2) \cong \mathbb{Z}\oplus\mathbb{Z}$$

Calculating the 2nd homology group $H_2$ The 2nd homology group $H_2(T^2;\mathbb{Z})$ represents the 2-dimensional "void" enclosed by the surface. Since the torus is a compact, oriented 2-manifold without a boundary, its top-dimensional homology group is isomorphic to $\mathbb{Z}$. In our model, the single 2-cell $f$ acts as a cycle because its boundary is zero, and there are no 3-cells to "fill" it.

$$H_2(T^2) \cong \mathbb{Z}$$

Higher-dimensional homology groups Because the torus is a 2-dimensional manifold, it contains no cells or structures of dimension 3 or higher. Consequently, all homology groups for $n \geq 3$ are trivial.

$$H_n(T^2) = 0 \text{ for } n \geq 3$$

Answer: ¡strong¿The homology groups of the torus $T^2$ are $H_0(T^2) \cong \mathbb{Z}$, $H_1(T^2) \cong \mathbb{Z}^2$, $H_2(T^2) \cong \mathbb{Z}$, and $H_n(T^2) = 0$ for $n \geq 3$.

**Describe relative homology.**

***Relative homology*** measures the homology of a space ***relative to a subspace***—intuitively, it captures the holes of $X$ that are "not already present in $A$".

Let's go step by step.

—

1. **Setup**
   Let $A \subset X$ be a subspace of a topological space $X$.
   Relative homology assigns groups

$$H_n(X, A)$$

   called the **relative homology groups** of the pair $(X, A)$.
   They measure how $X$ differs from $A$ topologically.
   —

2. **Definition using chains**
   In singular homology, we start with the chain groups:

$$C_n(X)$$

   generated by singular (n)-simplices in (X).
   Similarly we have

$$C_n(A)$$

   generated by simplices lying in $A$.
   Since $A \subset X$, we have

$$C_n(A) \subset C_n(X).$$

   The **relative chain group** is the quotient

$$C_n(X, A) = C_n(X)/C_n(A).$$

   So chains in $A$ are treated as **zero**.
   The boundary operator descends to this quotient, giving a chain complex

$$\cdots \to C_{n+1}(X, A) \to C_n(X, A) \to C_{n-1}(X, A) \to \dots$$

   Then

$$H_n(X, A) = \frac{\ker(\partial_n : C_n(X, A) \to C_{n-1}(X, A))}{\operatorname{im}(\partial_{n+1})}.$$

   —

3. **Geometric interpretation**
   Relative cycles are chains in $X$ whose boundary lies in $A$.
   So:
   * cycles may **end on** $A$ * boundaries are those filled inside $A$
   Thus relative homology detects features of $A$ that **become trivial once** $A$ **is allowed as a boundary**.
   —

4. **Example**
   Consider the disk and its boundary:

$$X = D^2, \qquad A = S^1.$$

   Here $S^1$ is the boundary of the disk.
   Key result:

$$H_2(D^2, S^1) \cong \mathbb{Z}.$$

   Interpretation:
   * The disk itself represents a **relative 2-cycle** * Its boundary lies in $S^1$
   So it is allowed in relative homology.
   Note that:

$$H_2(D^2) = 0$$

   but relative homology detects the 2-dimensional region.
   The boundary circle is the familiar space Circle.
   —

5. **Long exact sequence of a pair**
   Relative homology fits into a fundamental structure:
   *** Long exact sequence

$$\cdots \to H_n(A) \to H_n(X) \to H_n(X, A) \to H_{n-1}(A) \to H_{n-1}(X) \to \cdots$$

   This sequence shows how the topology of $A$, $X$, and the pair $(X, A)$ interact.
   It is one of the most important tools in algebraic topology.
   —

6. **Intuition**
   Relative homology answers the question:
   **What new holes appear when we enlarge $A$ to $X$?**
   or equivalently
   **Which cycles in $X$ become trivial once $A$ is allowed as a boundary?**
   —

7. **Special cases**
   If $A = \emptyset$

$$H_n(X, \emptyset) = H_n(X).$$

   So ordinary homology is a special case.
   —
   If $A = X$

$$H_n(X, X) = 0.$$

   Because everything becomes trivial.
   —

8. **Important applications**
   Relative homology is used in:

**excision theorem**
**Mayer–Vietoris sequence**
**Poincaré–Lefschetz duality**
manifolds with boundary
CW complex computations

For example, manifolds with boundary use

$$H_n(M, \partial M).$$

—

Key intuition

Relative homology studies **chains in $X$ whose boundaries lie in $A$**.

So it measures the topology of $X$ **relative to the part $A$ that we treat as negligible**.

—

If you'd like, I can also explain:

• the **disk/sphere example in detail** • the **exact meaning of relative cycles** • the **excision theorem (which makes relative homology powerful)** • how relative homology appears in **degree theory and manifolds**.

**Describe the Excision Theorem**

The **excision theorem** is one of the central results in relative homology. It tells us that ***removing a suitable subset from both $X$ and $A$ does not change the relative homology***. This makes relative homology extremely powerful because it lets us simplify spaces without affecting the answer.

—

1. **Statement of the excision theorem**
   Let $A \subset X$, and suppose $U \subset A$ is a subset whose closure lies in the interior of $A$:
   $$\overline{U} \subset \operatorname{int}(A).$$
   Then the inclusion
   $$(X - U, A - U) \hookrightarrow (X, A)$$
   induces isomorphisms
   $$H_n(X - U, A - U) \cong H_n(X, A)$$
   for all $n$.
   —
2. **Meaning of the theorem**
   We can **cut out** a region $U$ that lies completely inside $A$ without affecting the relative homology.
   So relative homology depends only on the portion of $X$ **near the boundary between $X$ and $A$**.
   Everything deep inside $A$ can be removed.
   —
3. **Intuition**
   Relative homology measures chains in $X$ whose boundaries lie in $A$.
   If a region $U$ sits entirely inside $A$:
   - any chains inside $U$ are already considered trivial
   - removing them does not affect the homology
     Thus the topology inside $A$ is irrelevant.
     Only the **interaction between $X$ and $A$** matters.

   —
4. **Why this theorem is useful**
   Excision allows us to **shrink complicated regions** to simpler ones.
   Many important computations in algebraic topology depend on it, including:
   - homology of spheres
   - CW complex homology
   - Mayer–Vietoris sequence
   - manifolds with boundary

   —
5. **Classic example: computing sphere homology**
   Consider the pair $(D^n, S^{n-1})$ where $D^n$ is a disk and $S^{n-1}$ its boundary.
   Using excision, one shows

$$H_k(D^n, S^{n-1}) = \begin{cases} \mathbb{Z} & k = n \\ 0 & k \neq n \end{cases}$$

This calculation helps prove the homology of spheres like the familiar boundary of a disk, the Circle.

—

6. **Geometric picture**

   Imagine:

   - $A$ is a thick region inside $X$
   - $U$ is a smaller piece deep inside $A$

   Excision says:

   we can carve out $U$ completely, and the relative homology stays the same.

   So the interior of $A$ becomes irrelevant.

   Only the **interface between $X$ and $A$** carries homological information.

   —

7. **Why excision is surprising**

   Ordinary homology **does not allow this**.

   If you remove a subset from a space, its homology usually changes.

   But relative homology is built so that:

   1. chains in $A$ are ignored
   2. therefore parts deep inside $A$ do not matter

   This special feature is exactly what excision formalizes.

   —

8. **Key idea behind the proof** *intuition*

   The proof relies on **subdividing singular simplices**.

   After enough subdivision:

   * every simplex lies either inside $X - U$ or inside $A$

   So chains can be replaced by equivalent chains that avoid $U$.

   Thus removing $U$ does not change the homology.

   —

9. **Key takeaway**

   The excision theorem says:

   If a subset lies entirely inside $A$, we can remove it from both $X$ and $A$ without changing the relative homology.

   This allows complicated spaces to be **simplified locally**, which is why excision is one of the most powerful tools in algebraic topology.

—

If you'd like, I can also show:

- **How excision leads to the computation of sphere homology**
- **Why $H_n(D^n, S^{n-1}) = \mathbb{Z}$**
- **How excision works inside the proof of Mayer–Vietoris**
- **A geometric picture of excision using chains and simplices**

These are usually the next big steps in mastering relative homology.